Campus WLAN Optimization

Using Access Point Placement and Network Segmentation

SRM Institute of Science and Technology

Five-member academic capstone team
Cisco Packet Tracer 9.0.1.0858

Campus scenario and problem statement

Floor	Functional zone	Representative clients
Ground	Classrooms	PC0, Students
Ground	Computer laboratory	PC1, Students; PC2, IoT proxy
Ground	Common area	Laptop0, Students
First	Library	PC3, Faculty; Laptop1, Guests
First	Faculty and administration	PC4, Administration; PC5, Management

Simulation assumption: eight campus endpoints, two APs and one external test host.

A shared client LAN exposes departmental resources. AP placement and channel choices need explicit planning.

Project objectives

1. Build two functioning and comparable campus topologies.
2. Retain distinct core, distribution and access layers.
3. Segment six roles and enforce permitted access.
4. Verify routing, DHCP and basic wireless forwarding.
5. Report observed outcomes and simulator limits.

Technical literature and validation gap

Verified reference	Relevant guidance	Project application
Cisco campus design [1]	Separate access, policy and transport roles	Explicit three-tier academic constraint
Cisco site survey [2]	Validate coverage and capacity in the environment	No measured RF-coverage claim
Cisco RF channels [3]	Avoid overlapping adjacent channel choices	Configured channel plan, not measured gain
RFC 2131 [4]	DHCP addressing and relay behavior	Verify leases across the routed core

Validation gap: production guidance does not prove feature behavior in a Packet Tracer model.

[1–4] Full verified URLs and reference candidates appear in the report evidence package.

Baseline and optimized systems

Aspect	Baseline	Optimized
Device and client population	Same 16 devices	Same 16 devices
Campus client connectivity	Shared VLAN 1 and /24	Six role VLANs and /24s
Departmental policy	No departmental ACLs	Distribution SVI ACLs
AP arrangement	Central clustering, channels 1 / 1	Spread by zone, channels 1 / 6
Wireless technology	Standalone AP-PT, WPA2-PSK	Same AP and client models
External test path	Distribution, core, edge	Same routed path

Both reference networks function. The comparison exposes policy differences without creating faults.

Project methodology

01 Audit files and preserve historical evidence

02 Build and verify the functioning flat baseline

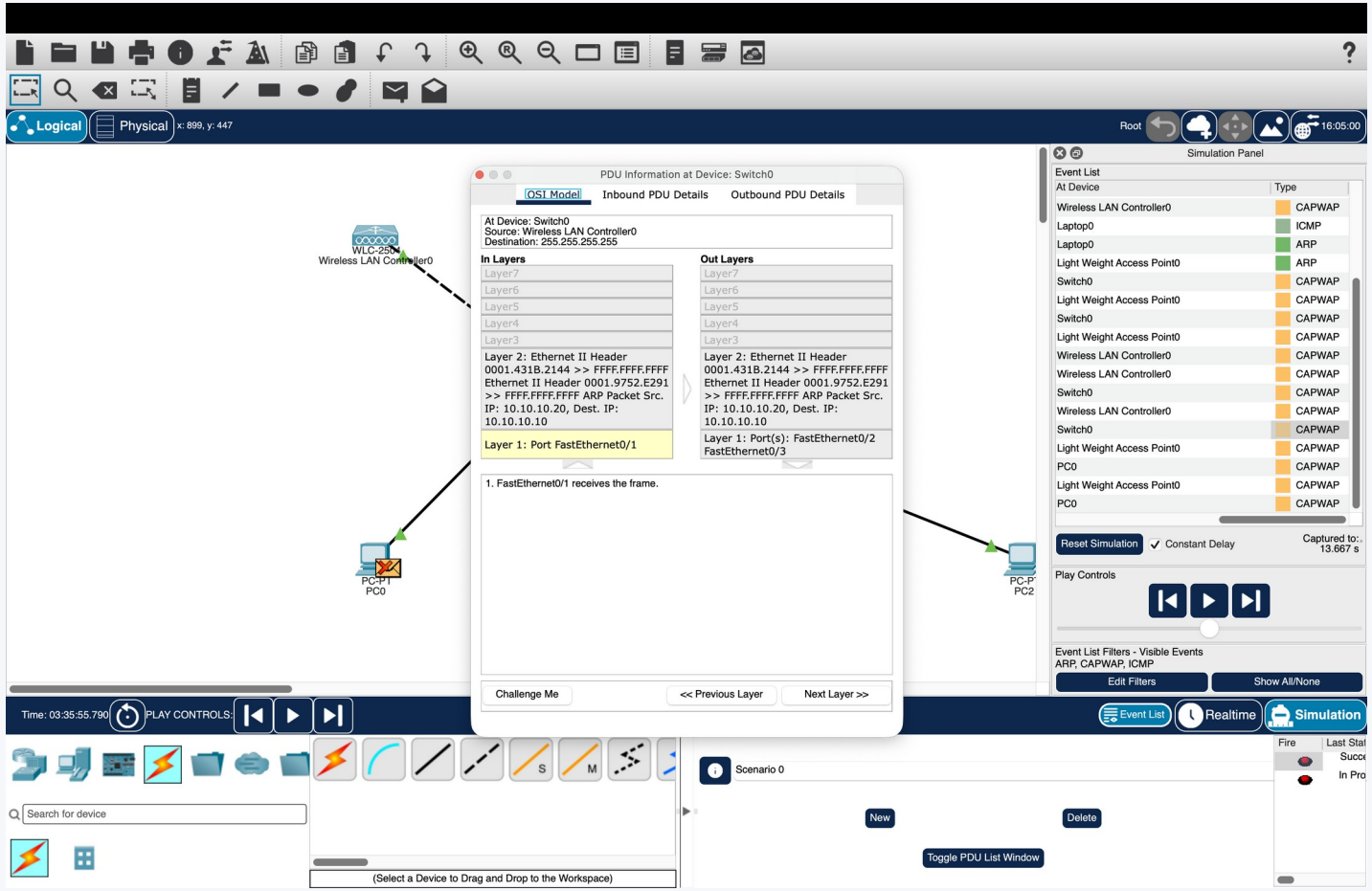
03 Apply role VLANs, SVI routing, DHCP relay and ACLs

04 Use standalone APs after the WLC forwarding failure

05 Run permitted and denied probes with positive controls

06 Save, reopen and connect every result to its evidence

Root cause analysis and wireless fallback



Observed in Test H:

Client associates but data fails.

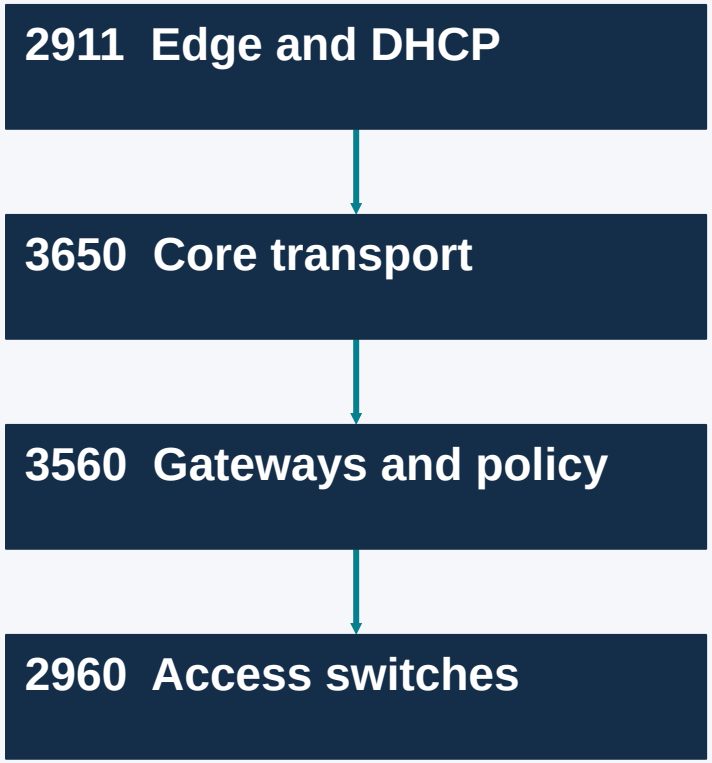
Fresh client traffic appears in VLAN 99.

The intended VLAN 10 target is absent.

Root cause remains unresolved.

Decision: preserve the failed WLC result and demonstrate basic wireless using tested standalone APs.

Three-tier architecture and VLAN design

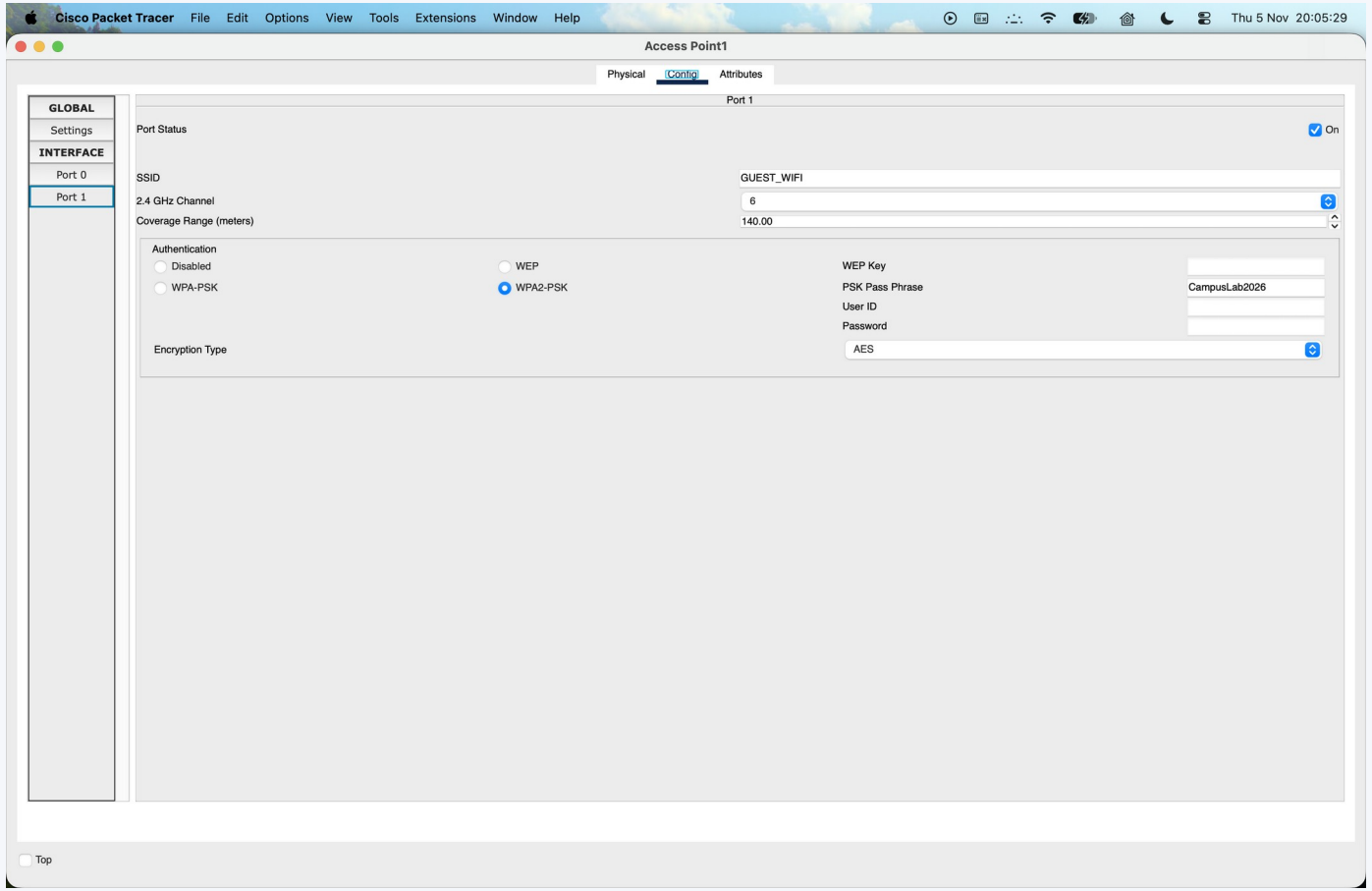


VLAN	Role	Subnet
10	Students	10.10.10.0/24
20	Faculty	10.10.20.0/24
30	Administration	10.10.30.0/24
40	Guests	10.10.40.0/24
50	IoT proxy	10.10.50.0/24
99	Management	10.10.99.0/24

Distribution owns .1 gateways.
Access uplinks use 802.1Q.
Core links use routed /30s.

```
interface vlan 10
 ip helper-address 10.255.0.1
 ip access-group STUDENTS_IN in
```

AP placement and wireless strategy



Two APs in both scenarios

STUDENT_WIFI on access VLAN 10

GUEST_WIFI on access VLAN 40

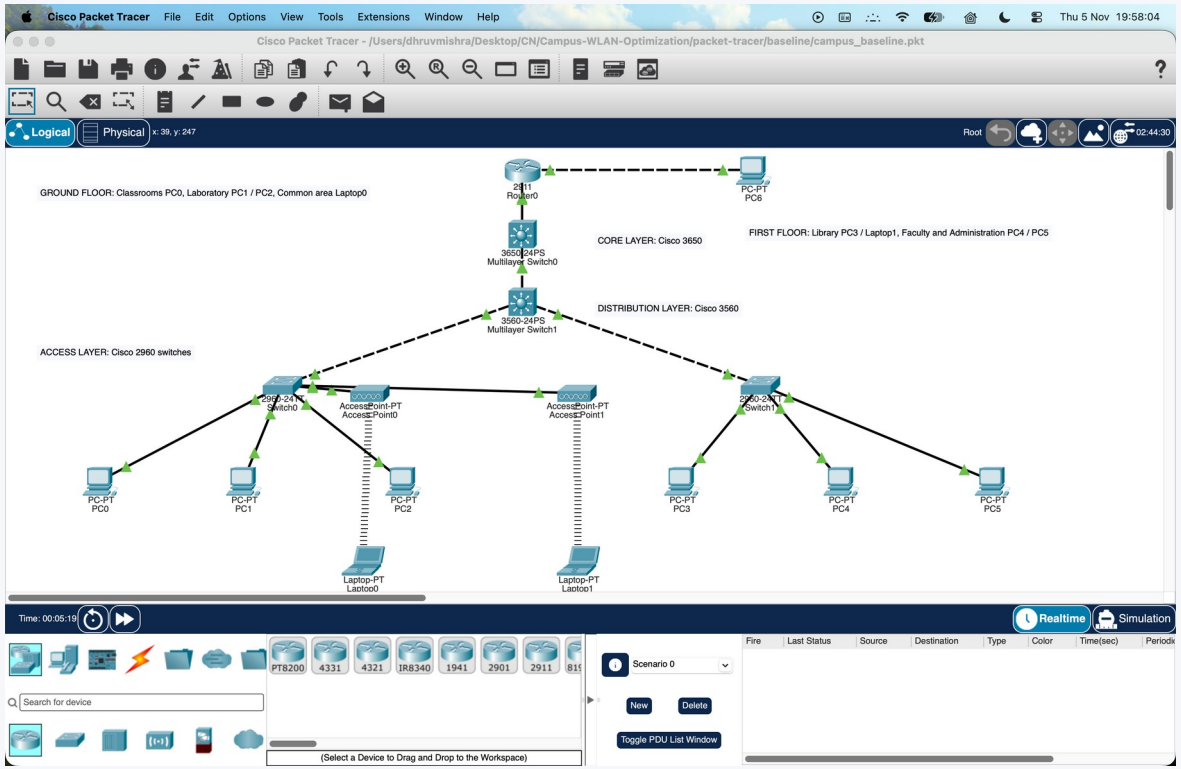
WPA2-PSK with AES in the lab

Channels 1 and 6 in the optimized copy

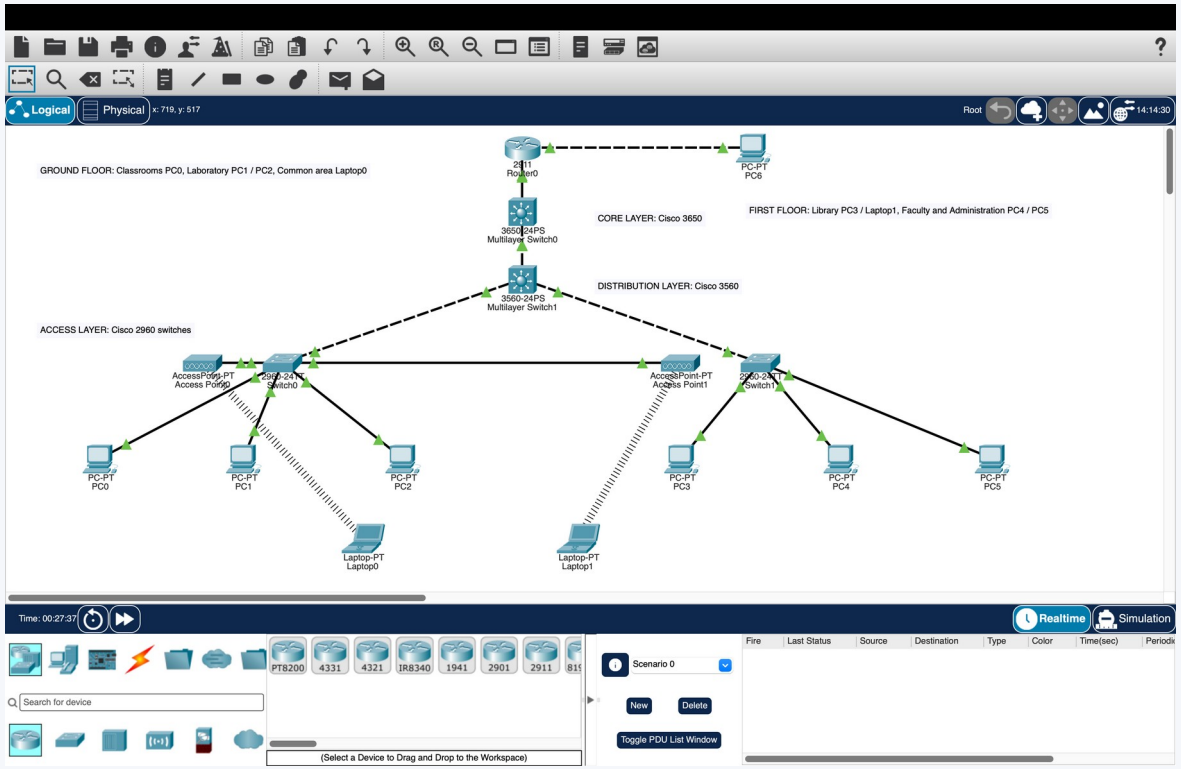
Zone placement is illustrative. No coverage percentage, RF capacity or roaming improvement was measured.

Actual Packet Tracer implementation

Baseline



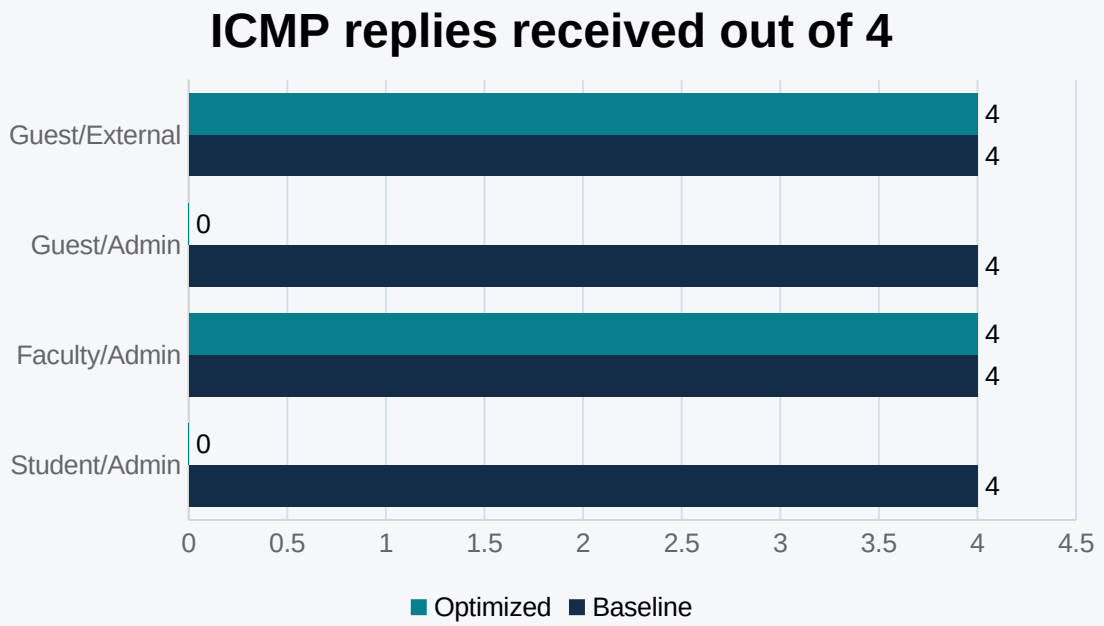
Optimized



Native .pkt files, configuration transcripts, DHCP bindings, ACL counters and screenshots accompany this deck.

Verified functional comparison

Four-packet probe	Baseline	Optimized
Student to Administration	4/4	0/4
Faculty to Administration	4/4	4/4
Guest to Administration	4/4	0/4
Guest to external host	4/4	4/4



Zero replies on unauthorized flows indicate intended policy denial, supported by allowed controls and ACL evidence.

No throughput or RF coverage claim. These are completed functional probes, not a packet-loss benchmark.

Conclusion, limitations and future scope

The campus demonstrator implements distinct network tiers and role-based policy.

Standalone wireless provides a defensible basic WLAN demonstration.

WLC VLAN forwarding remains unresolved in its separate historical fixture.

RF coverage, congestion, roaming and throughput require stronger measurement tools.

Next: apply the supplied report template, add team identities and rehearse the saved-file demonstration.